

Lab Report: Azure Global Hub-and-Spoke Networking with Transitive Routing

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Objective: To design and implement a scalable Hub-and-Spoke network topology across two Azure regions (East US and West US), enabling centralized management and transitive communication between spokes using a custom Network Virtual Appliance (NVA).

1. Executive Summary

This lab demonstrates the configuration of a complex Azure environment featuring Global VNet Peering, Private DNS Zones, and User-Defined Routes (UDR). The core achievement was bypassing Azure's non-transitive peering limitation by utilizing a Windows Server VM as an NVA to route traffic between disparate spokes.

2. Network Architecture & Topology

Component	Resource Name	Region	Address Space
Hub VNet	vnet-hub-eastus	East US	10.0.0.0/16
Spoke 1 VNet	vnet-spoke1-eastus	East US	10.1.0.0/16
Spoke 2 VNet	vnet-spoke2-westus	West US	10.2.0.0/16

3. Implementation Steps

Phase 1: Foundation and Connectivity

- VNet Peering:** Established Global VNet Peering between the Hub and both Spokes. Enabled "Allow Forwarded Traffic" on all peering links to support NVA functionality.
- DNS Configuration:** Implemented a Private DNS Zone (internal.corp) linked to all three VNets with auto-registration enabled to allow name resolution regardless of region.

Phase 2: Custom Routing (Transitive Traffic)

- NVA Configuration:** Enabled **IP Forwarding** on the Hub VM's Network Interface (NIC) in the Azure Portal.
- Guest OS Routing:** Modified the Windows Registry (IpEnableRouter = 1) inside the Hub VM to allow it to forward packets between subnets.

- **Route Tables (UDR):** Created regional Route Tables in East US and West US. Added a route for 10.0.0.0/8 with the "Next Hop" set to the Hub VM's private IP (10.0.1.4).

Phase 3: Validation and Troubleshooting

- **Host-Level Security:** Identified that ICMP (Ping) was blocked by the Windows Defender Firewall. Resolved by adding an inbound rule for ICMPv4 on all Spoke VMs.
- **Network Watcher:** Utilized "Next Hop" and "IP Flow Verify" to confirm packets were hitting the Hub VM instead of the default Azure system routes.

4. Key Takeaways

- **Peering is Not Transitive:** By default, Spoke A cannot talk to Spoke B through a Hub. A routing mechanism (NVA or Azure Firewall) must be manually introduced.
- **Global Latency:** Observed an increase in latency (~70ms) when communicating between East US and West US compared to local region traffic (~1ms).
- **Layered Troubleshooting:** Connectivity requires three green lights: Azure Peering, Subnet Routing (UDR), and Guest OS Firewall settings.

5. Cleanup Procedure

To optimize cost management, the entire Resource Group rg-hub-spoke-lab was deleted, terminating all compute and networking assets simultaneously.